Verasonics®
The leader in Research Ultrasound™

Vantage® *NXT* Research Ultrasound Systems

FastSave Benchmark Results

Verasonics Inc. | 11335 NE 122nd Way, Ste 100, Kirkland, WA 98034 | www.verasonics.com
©Verasonics Inc. 2001- 2025. Verasonics Confidential.

Table of Contents

1. FastSave Benchmark results *2*

2. Test environment *2*

3. Drive Benchmarking (FIO) *3*

3.1 Linux: Per-Drive FIO Results 3

3.2 Windows: Per-Drive FIO Results 5

3.3 Summary 6

4. Application-Level Acquisition Benchmarks *7*

4.1 Acquisition Validation Images 8

4.2 Observations 8

5. Notes, assumptions, and recommended additional detail *9*

6. Conclusion *9*

1. FastSave Benchmark results

These results summarize in-house Verasonics testing conducted on October 29, 2025. Results are intended as reference only; actual performance depends on hardware, firmware, OS configuration, filesystem, and test parameters.

2. Test environment

Item	Configuration
Verasonics host controller	Expandable (Dell workstation used for testing)
Workstation model	Dell Precision 5860 XL
CPU	Intel Xeon W7-2475X (2.59 GHz)
RAM	64 GB DDR5
PCIe slot	PCIe Gen4 ×16
Windows	Windows 11 IoT Enterprise LTSC 24H2
Linux	Ubuntu 22.04, kernel 5.15
NVMe controller card	Dell Ultra-Speed Drive Quad FH Bracket (Manufacturer part 7YJH0 / Dell part 400-BPJP)
NVMe storage (test units)	4 × 1 TB WD_BLACK (SN850X and SN8100 variants tested)
Verasonics hardware	Vantage NXT, 256-channel HF
Vantage software version	2.2.0
fio version	3.38

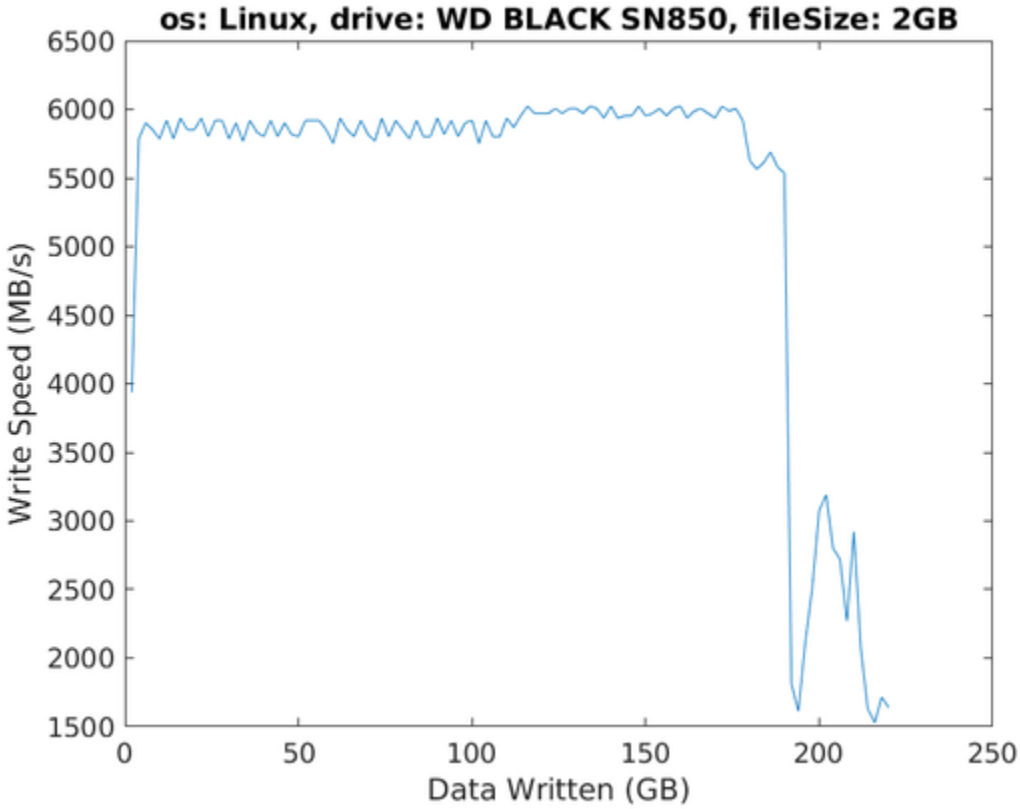
Notes:

- The NVMe models tested include WD SN850X (PCIe Gen4) and WD SN8100 (PCIe Gen5-rated).
- Filesystem and mount options strongly affect sustained write performance (see Recommendations below).

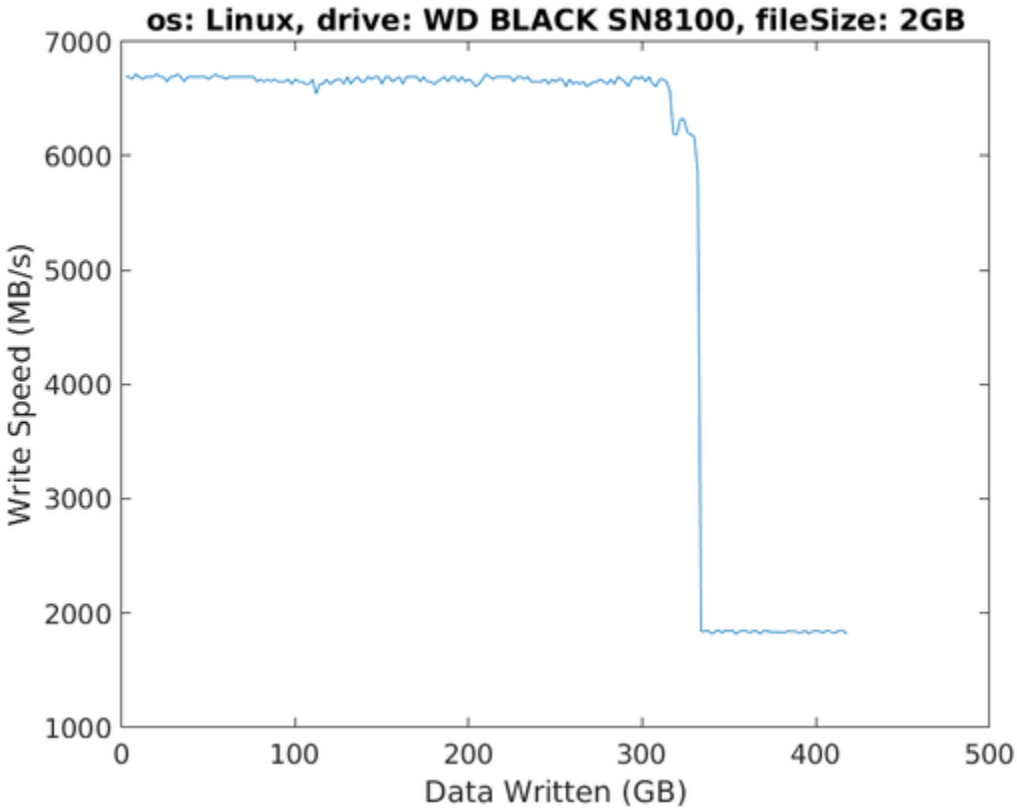
3. Drive Benchmarking (FIO)

The following figures show sequential write benchmarks (per-drive) performed with FIO.

3.1 Linux: Per-Drive FIO Results



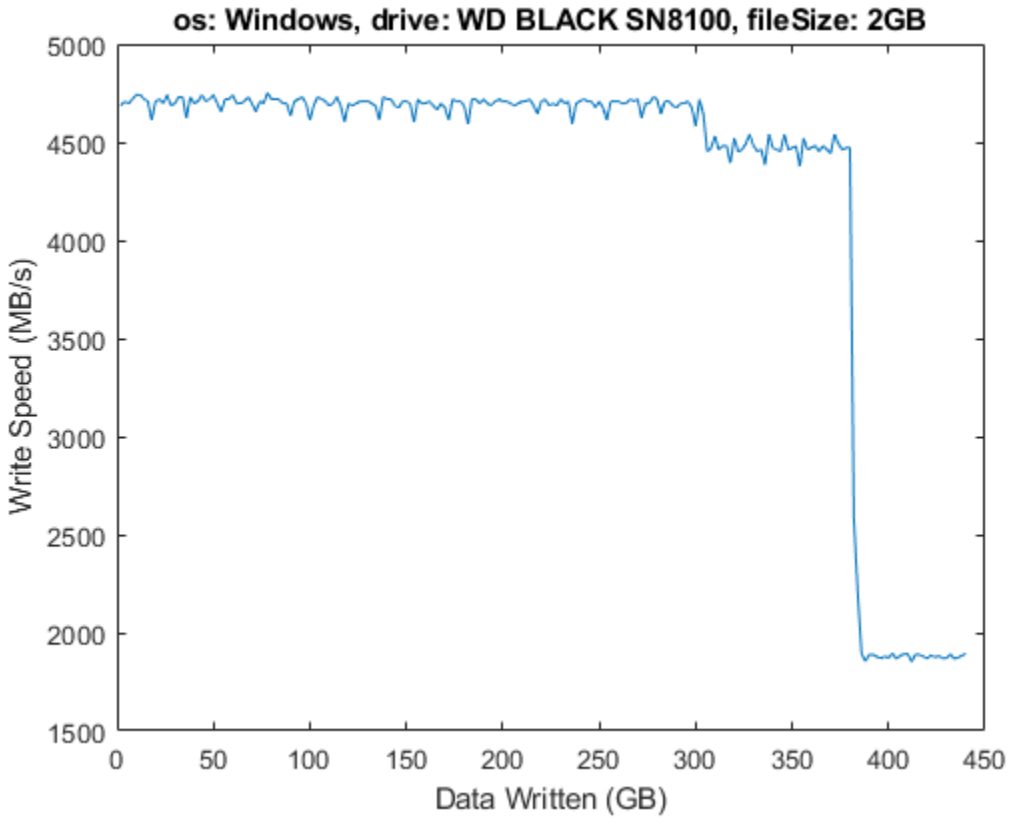
Linux fio, SN850X 2GB job



Linux fio, SN8100 2GB job

3.2 Windows: Per-Drive FIO Results

Windows fio, SN850X 2GB job



Windows fio, SN8100 2GB job

3.3 Summary

- The SN8100 (Gen5-rated) shows higher peak sequential write capability compared with the SN850X (Gen4). In our test system the SN8100's peak performance was constrained by the PCIe Gen4 host interface.
- Observed Windows throughput is lower than Linux in these tests; possible contributors include filesystem differences (NTFS vs ext4), OS caching/policies, driver differences, and background services.

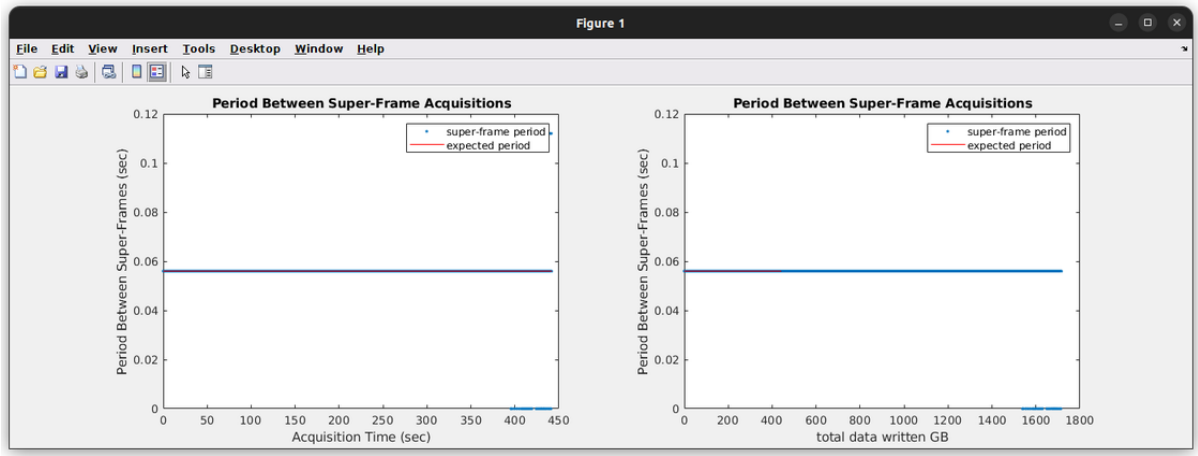
4. Application-Level Acquisition Benchmarks

The following acquisition tests were performed using the supplied example acquisition script (L22-4v FlashAngles variant) and FastSave writing data across four NVMe devices.

Metric	Observed value
Total available storage	4 TB
Super-frame size	0.22 GB
Acquisition rate (raw frames)	890 fps
Super-frame rate	18 super-frames / s
Average data throughput (sustained)	3.9 GB/s
Max number of files written	18,357
Maximum acquisition duration (at sustained rate)	1,028 s (~17.1 min)

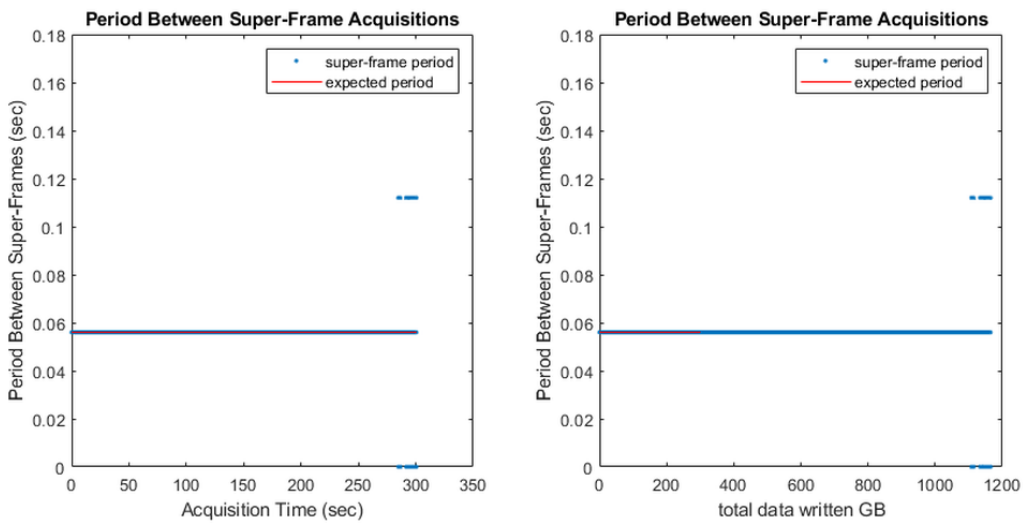
4.1 Acquisition Validation Images

Linux (4 × SN850X):



Acquisition validation - Linux

Windows (4 × SN850X):



Acquisition validation - Windows

4.2 Observations

- On Linux, the L22-4v FlashAngles example sustained ~3.9 GB/s continuous write for >6 minutes (~1.5 TB) without dropped frames under the tested configuration.
- On Windows, the same acquisition sustained approximately 3.9 GB/s for just under 5 minutes (~1.2 TB) before reaching limitations observed in the test (no dropped frames reported within the sustained window).

5. Notes, assumptions, and recommended additional detail

- Average data throughput is computed as the total stored data divide by duration of acquisition marked by timetags in the first to the last frame before a dropped frame.
- Thermal throttling most likely does not affect throughput. Though the data is not presented here, the tempreature of the NVMe drives was monitored and never appeared to increase above a throtteling threshold.

6. Conclusion

The supplied benchmarks show that, with the tested hardware and Linux configuration, FastSave can sustain multi-GB/s writes across four NVMe devices for several minutes without dropped frames. Windows performance in these tests was lower; further investigation could help uncover the limiting factor and possible mitigations.

This document is intended for use by Verasonics' customers only.
Do not duplicate or distribute without permission.